

# Conditionals, Couplings and Toric Geometry: Kernels, the Coupling Nerve, Cumulants and a Fukaya Programme

Revised draft 4

Vinay K. Awasthi  
Department of Applied Mathematics and Statistics  
Johns Hopkins University\*

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## Abstract

We organize the conditional and coupling structure of the finite distribution monad  $D$  into four layers and separate what is proved from what is conjectural.

- (1) *The base.* Markov kernels are the 1-cells of the Kleisli category, and couplings are 2-cells whose vertical composition is the gluing lemma. The natural whiskerings are not functorial, so the couplings do not form a 2-category. Their image under the cost map does: a strict, graded, Lawvere-enriched 2-category.
- (2) *The fibre.* The base-relative posterior tower is a functor on the Kleisli category. Its category of elements is a discrete opfibration over it, and every grade of the shadow bounds the distance between transported contexts at every level of the tower.
- (3) *The coupling nerve.* Couplings of all arities form the simplicial object  $D(Y^{\bullet+1})$ , with marginalization as faces. Every 2-horn fills. The fillers form a product of coupling polytopes, and the maximum-entropy filler is the one whose third face is the vertical composite. For  $n \geq 3$  the faces of an  $n$ -horn form a cyclic cover, so filling can fail. The failure is measured by the contextual fraction, and  $D(Y^{\bullet+1})$  is not a Kan complex, although  $Y^{\bullet+1}$  is.
- (4) *Linearization.* Centred couplings are second mixed cumulants, and whiskering defects are the Boolean cumulants of whiskering viewed as a noncommutative expectation. Into the square-zero cone of the circulation ideal, whiskering is an  $A_\infty$ -functor with exactly two components. Its cohomology keeps only the marginals, and the coupling data are homotopical.

On top of these layers sits the geometry. Coupling polytopes of nondegenerate marginals are Delzant, so they are moment polytopes of toric manifolds whose Kähler class is the product coupling and whose canonical metric restricts to the Fisher metric. Their walls and chambers, monotone cases, Floer theory and Bridgeland stability conditions are treated as before. A new section takes up the five problems the previous draft left open. The parity horns have contextual fraction  $1 - 2^{2-n}$  for every  $n$ . Whiskering extends to a multi-object  $A_\infty$ -functor that receives the 2-skeleton of the nerve. Wall-crossing is variation of GIT. Central charges glue continuously across the pentagon–hexagon wall. The Floer-nontrivial fibres lie over the leximin coupling, not over the product coupling. A further section connects the obstruction tower: every horn obstruction is the secondary obstruction of an encoded level-one claim, equal to twice the contextual fraction when all compatible families are admissible, and the Fisher metric bounds the obstructions from above. Conjectures are stated with correct types, and a companion paper [4] computes every construction explicitly.

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\*Verification scripts are listed in [section 13](#); the worked examples are in the companion [4].

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## 1 Introduction

Two structures sit on top of the finite distribution monad  $D$ . The base-relative posterior tower [2] fixes an outcome base  $B$  and solves a *strict integrability* problem: it carries a mixed distributive law satisfying all four Beck axioms. The conditional/coupling structure [1] lets the base vary along Markov kernels and compares kernels by couplings: it solves a *transport* problem. This paper assembles them, together with a simplicial and a homotopical layer, into one framework.

| Layer             | Objects and structure                                     | Status                                                                    |
|-------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| I. Base           | finite sets, kernels, coupling 2-cells; the graded shadow | strict 2-category on grades only (section 2)                              |
| II. Fibre         | the tower over each base; transport along kernels         | discrete opfibration; grades act on fibres (section 3)                    |
| III. Nerve        | $D(Y^{\bullet+1})$ ; horns, fillers, boundary fibres      | 2-horns fill; higher horns obstructed by contextuality (section 4)        |
| IV. Linearization | circulations, cumulants, cones                            | whiskering is a two-component $A_\infty$ -functor into a cone (section 5) |
| Geometry          | coupling polytopes as moment polytopes; stability         | proved in the nondegenerate case (sections 6 to 8)                        |
| Programme         | families over chambers; Fukaya-type functor               | conjectural (section 9)                                                   |

The layers interlock. A 2-cell is a 1-simplex of the nerve, and vertical composition is the canonical filler of an inner 2-horn. The tangent spaces of the filler polytopes are circulations, and those are where the cumulants of layer IV live. The tower of layer II is transported along the 1-cells of layer I, and the grades of layer I bound the transport. The polytopes of layer III are, in the nondegenerate case, moment polytopes, and that is where the geometry enters. What does *not* hold is equally important. The couplings themselves do not form a 2-category, the nerve is not Kan, and the toric picture stops at boundary fibres of 3-way tables.

## 2 Layer I: kernels, couplings and the graded shadow

$D$  is the finitely supported distribution monad on **Set** and  $\mathbf{Kl}(D)$  its Kleisli category. A 1-cell  $k: B \rightsquigarrow B'$  is a Markov kernel  $k: B \rightarrow D(B')$ , acting on states by  $k^\#(p)(b') = \sum_b p(b)k(b)(b')$ . The results of this section are proved in [1, §7]; we restate them for reference.

**Proposition 2.1** (1-cells [1, Props. 7.2, 7.3]).  *$k^\#$  is affine,  $(h \circ k)^\# = h^\#k^\#$ ,  $k^\# \circ \mu = \mu \circ D(k^\#)$ , and  $\mathrm{TV}(k^\#p, k^\#q) \leq \mathrm{TV}(p, q)$ . Attribution sets are transported along  $k$ , and precision can only be lost.*

**Definition 2.2** (Coupling 2-cells). For 1-cells  $f, g: X \rightsquigarrow Y$ , a coupling 2-cell  $\alpha: f \Rightarrow g$  is a kernel  $\alpha: X \rightarrow D(Y \times Y)$  with marginals  $f(x)$  and  $g(x)$ . Its cost is  $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$ ; the identity is the diagonal coupling; the vertical composite glues along the common marginal,  $(\beta \cdot \alpha)(x)(y, y'') = \sum_{y'} \alpha(x)(y, y')\beta(x)(y', y'')/g(x)(y')$ .

**Theorem 2.3** (Witnesses and shadow [1, Thms. 7.6, 7.8]).

- (i) *For fixed  $X, Y$ , 1-cells and coupling 2-cells form a category under gluing, strictly associative and unital. Cost is subadditive and  $\min_\alpha c(\alpha) = d(f, g) = \max_x \mathrm{TV}(f(x), g(x))$ .*
- (ii) *Whiskering by a deterministic precomposition is strictly functorial. Whiskering by a stochastic precomposition (mixing couplings) or by a postcomposition (copying on equal pairs, independent on unequal pairs) preserves identities, marginals and the cost bound, but is not functorial.*
- (iii) *Graded 2-cells  $f \Rightarrow_\varepsilon g$ ,  $\varepsilon \geq d(f, g)$ , with grades adding, form a strict 2-category on kernel 1-cells, and  $d$  makes  $\mathbf{Kl}(D)$  a Lawvere-enriched category.*

**Remark 2.4** (Witness layer and shadow). The cost map  $\alpha \mapsto c(\alpha)$  sends the witness layer onto the graded shadow. It preserves identities and sends composites to grades no larger than the sum. The

shadow is an image of the witness layer, not a substructure of it, and it is the only strict 2-category in sight: with couplings as 2-cells, (ii) rules out a 2-category or a bicategory with these whiskerings. The polytopes  $\text{Cpl}(f(x), g(x))$  are the spaces of 2-cells, not further cells; their geometry is the subject of [sections 4 and 6](#).

### 3 Layer II: the tower as a fibration

The base-relative tower over  $B$  has context algebra  $A_B = \text{DB} \times \text{D}^2 B \times \text{D}^3 B \times \text{Sh}(B)$ , reader comonad  $\mathbb{C}^B(Y) = Y \times A_B$  and strict law  $\lambda^B$  [2, §9]. With  $\text{Hyp} = \Theta \times \text{D}(-)$ , a kernel  $k: B \rightarrow \text{D}(B')$  acts on hypotheses by  $\text{Hyp}(k)(\theta, \rho) = (\theta, k^\# \rho)$ , and we put  $A_k = k^\# \times \text{D}(k^\#) \times \text{D}^2(k^\#) \times \text{D}(\text{Hyp}k)$ .

**Theorem 3.1** (The tower is a functor on the Kleisli category [2, Prop. 9.14, Cor. 9.15, Prop. 9.16]).

- (i)  $A_k$  is a morphism of  $\text{D}$ -algebras, preserves coherent contexts, commutes with the canonical section  $\sigma$ , and  $A_{h \circ k} = A_h A_k$ ; for deterministic  $k$  it is the earlier base change.
- (ii)  $\text{id} \times A_k$  is a comonad morphism  $\mathbb{C}^B \Rightarrow \mathbb{C}^{B'}$  commuting with the laws  $\lambda^B, \lambda^{B'}$ .
- (iii) Admissibility data, coherent lifts and the effective sets are transported; certificates pull back along  $f' \mapsto k \triangleright f'$ ; failure levels can only rise and the secondary obstruction can only fall.
- (iv) Conditioning on an outcome observed after the channel equals conditioning on the pulled-back predicate  $k \triangleright 1_{e'}$ , then changing base.

**Definition 3.2** (The total category). Let  $\mathcal{E}$  be the category whose objects are pairs  $(B, z)$  with  $B$  a finite set and  $z \in A_B^{\text{coh}}$  a coherent context, and whose morphisms  $(B, z) \rightarrow (B', z')$  are kernels  $k$  with  $A_k(z) = z'$ . Let  $\pi: \mathcal{E} \rightarrow \text{Kl}(\text{D})$  forget  $z$ .

**Proposition 3.3** (Split opfibration).  $\pi$  is a discrete opfibration, and in particular a split opfibration. The cocartesian lift of  $k: B \rightarrow B'$  at  $(B, z)$  is  $k: (B, z) \rightarrow (B', A_k z)$ . Through  $\text{id} \times A_k$ , the comonads  $\mathbb{C}^B$  and the laws  $\lambda^B$  are carried along the lifts.

*Proof.*  $\mathcal{E}$  is the category of elements of the functor  $B \mapsto A_B^{\text{coh}}$ ,  $k \mapsto A_k$ , which is a functor by [theorem 3.1\(i\)](#); categories of elements of set-valued functors are discrete opfibrations. The statement about comonads and laws is [theorem 3.1\(ii\)](#).  $\square$

**Proposition 3.4** (Grades act on fibres). Let  $f, g: B \rightarrow \text{D}(B')$  and let  $z = (\rho, \Pi, \Omega, \Sigma)$  be a coherent context. Put  $\text{TV}$  on  $\text{DB}'$ , the Kantorovich metric  $W_1$  over  $\text{TV}$  on  $\text{D}^2 B'$ , the Kantorovich metric over that on  $\text{D}^3 B'$ , and on hypotheses the metric that is  $\text{TV}$  of predictives for equal descriptors and 1 otherwise. Then every component of  $A_f(z)$  and  $A_g(z)$  is within  $d(f, g)$ . So a graded 2-cell  $f \Rightarrow_\varepsilon g$  gives  $\varepsilon$ -close transported contexts at every level.

*Proof.*  $\text{TV}(f^\# \rho, g^\# \rho) \leq \sum_b \rho(b) \text{TV}(f(b), g(b)) \leq d(f, g)$  by convexity. For  $\Pi$ , push  $\Pi$  forward along  $\rho \mapsto (f^\# \rho, g^\# \rho)$ . This is a coupling of  $\text{D}(f^\#) \Pi$  and  $\text{D}(g^\#) \Pi$  of cost  $\mathbb{E}_\Pi \text{TV}(f^\# \rho, g^\# \rho) \leq d(f, g)$ . Level three and the shadow are the same argument one level up. The companion verifies the bound in a worked example, where all three levels give  $\frac{2}{15} \leq \frac{1}{6}$  [4, §11].  $\square$

The proof of [theorem 3.1](#) uses one fact beyond naturality: Kleisli extension is a morphism of free  $\text{D}$ -algebras. All other components of  $A_k$  are pushforwards along functions. So the strict integrability of each fibre and the transport between fibres are compatible.

## 4 Layer III: the coupling nerve

**Definition 4.1** (The coupling nerve). For a finite set  $Y$ , let  $N_k(Y) = D(Y^{k+1})$ , with face maps  $d_i$  that marginalize out the  $i$ th coordinate and degeneracies  $s_i$  that push forward along the diagonal repeating it. Since  $Y^{\bullet+1}$  is a simplicial set (the Čech nerve of  $Y \rightarrow 1$ ) and  $D$  is a functor,  $N(Y) = D(Y^{\bullet+1})$  is a simplicial set. The relative version over a set  $X$  has  $k$ -simplices the kernels  $X \rightarrow D(Y_0 \times \cdots \times Y_k)$ , with all structure taken pointwise in  $x$ .

A 1-simplex of  $N(Y)$  is a coupling of its two vertices. A coupling 2-cell  $\alpha: f \Rightarrow g$  between kernels  $X \rightsquigarrow Y$  is an  $X$ -indexed 1-simplex with vertices  $f$  and  $g$ . A horn  $\Lambda_k^n$  in  $N$  is a compatible family of laws on the  $n$  faces other than the  $k$ th, and a filler is an  $n$ -simplex with those faces. In probabilistic terms a horn is an empirical model on the cover of  $\{0, \dots, n\}$  by those faces, and a filler is a global section [19].

**Theorem 4.2** (2-horns fill, and composition is the canonical filler). *Let  $p_{01}$  and  $p_{12}$  be compatible laws on  $Y_0 \times Y_1$  and  $Y_1 \times Y_2$  with common marginal  $p_1$ , the data of an inner horn  $\Lambda_1^2$ .*

- (i) *The fillers form the product, over  $x_1$  with  $p_1(x_1) > 0$ , of the transportation polytopes of the slices  $p(\cdot, x_1, \cdot)$  with marginals  $p_{01}(\cdot, x_1)$  and  $p_{12}(x_1, \cdot)$ . If every slice has full support and nondegenerate marginals, the filler polytope is Delzant.*
- (ii) *The gluing filler  $p^{\text{gl}} = p_{01} p_{12} / p_1$  is the unique filler of maximal entropy, and its face  $d_1$  is the vertical composite of the two couplings.*
- (iii) *Any filler differs from  $p^{\text{gl}}$ , slice by slice, by  $p_1(x_1) \text{Cov}(1_{\{x_0\}}, 1_{\{x_2\}} \mid x_1)$ , a circulation of the slice.*

*The outer horns  $\Lambda_0^2$  and  $\Lambda_2^2$  fill in the same way, by gluing over the shared vertex.*

*Proof.* (i) Each marginal constraint involves one value of  $x_1$ , so the constraints decouple into slices. Each slice is a two-way transportation polytope, Delzant under the stated conditions (theorem 6.4), and a product of Delzant polytopes is Delzant with respect to the product lattice. (ii)  $\log p^{\text{gl}}$  is a sum of a function of  $(x_0, x_1)$  and a function of  $(x_1, x_2)$ , so it is orthogonal to every tangent vector of the filler polytope. So  $p^{\text{gl}}$  is a critical point of the strictly concave entropy on the polytope, hence its unique maximizer. Equivalently,  $H(X_0 X_1 X_2) \leq H(X_0 X_1) + H(X_1 X_2) - H(X_1)$ , with equality exactly under conditional independence. Summing  $p^{\text{gl}}$  over  $x_1$  gives the gluing formula of definition 2.2. (iii) is a direct computation.  $\square$

So the 2-cells of layer I are the 1-simplices of the nerve, and their vertical composition is the canonical filling of inner 2-horns. The other fillers are conditional cumulants, which is the link to layer IV.

**Theorem 4.3** (Higher horns). *Let  $n \geq 3$ .*

- (i) *The  $n$  faces of  $\Lambda_k^n$ , viewed as subsets of  $\{0, \dots, n\}$ , form a cover that is not acyclic. For  $n = 2$  it is acyclic.*
- (ii) *Hence, for  $|Y| \geq 2$ , there are compatible horn data in  $N(Y)$  with no filler.*
- (iii) *For compatible horn data, the minimal unmodeled weight with the deterministic global assignments as sources, each face weighted  $1/n$ , equals the contextual fraction of the horn, and it vanishes exactly when the horn fills. For the parity horns on two outcomes (uniform on assignments of fixed parity on each face, with exactly one face odd) it is  $1 - 2^{2-n}$  for every  $n$ -horn.*

*Proof.* (i) Use Graham's reduction for acyclic hypergraphs [20]: repeatedly delete vertices in a single edge and edges contained in others; the hypergraph is acyclic if and only if this empties it. In  $\Lambda_k^n$  the vertex  $k$  lies in all  $n$  faces and every other vertex lies in  $n - 1$  of them. No face contains another, so for  $n \geq 3$  no reduction step applies. For  $n = 2$  the two faces share one vertex and reduce away. (ii) By Vorob'ev's theorem [31], every compatible family extends to a global law exactly on acyclic covers. Data on two outcomes embed into any  $Y$  with  $|Y| \geq 2$ . (iii) A filler is a global section, so this is [1, Thm. 7.12] applied to the horn's cover. For the parity horn on bits  $x_0, \dots, x_n$ , put  $\sigma = \sum_j x_j$  and let  $p_i \in \{0, 1\}$  be the parity of face  $i \neq k$ . A global assignment meets every face parity if and only if  $x_i = \sigma - p_i$  for all  $i \neq k$ ; then  $x_k$  is fixed by  $\sum_j x_j = \sigma$ , and this is consistent because  $\sum_i p_i = 1$ . So exactly two global assignments, one for each  $\sigma$ , are compatible with all faces, and they differ on every face. Every other deterministic source puts mass on an assignment of probability zero on some face. On each face the two compatible sources have distinct restrictions of probability  $2^{1-n}$ , so each can carry weight at most  $2^{1-n}$ ; the noncontextual fraction is  $2^{2-n}$ . Exactly verified certificates for  $n \leq 6$  and the count for  $n \leq 8$  are in `check_horns.n.py`.  $\square$

**Corollary 4.4** (The nerve is not Kan). *For  $|Y| \geq 2$ ,  $N(Y) = D(Y^{\bullet+1})$  fills every 2-horn but not every  $n$ -horn for  $n \geq 3$ . In particular it is not a Kan complex, although  $Y^{\bullet+1}$  is. The monad  $D$  does not preserve horn filling, because it does not turn compatible families over cyclic covers into global laws, and the obstruction is the contextual fraction.*

**Proposition 4.5** (Boundary fibres). *Prescribing all three faces of a 2-simplex, that is all three 2-margins of a 3-way table on  $Y_0 \times Y_1 \times Y_2$ , gives a planar 3-way transportation polytope of dimension  $(n_0 - 1)(n_1 - 1)(n_2 - 1)$  when it is nonempty and generic. It can be empty even for compatible faces, because the cover by the three edges of a triangle is cyclic.*

(i) *For  $n_0 = 2$  it is a capacitated two-way transportation polytope, whose constraint matrix is totally unimodular, so it is Delzant whenever it is simple.*

(ii) *Every rational polytope is a planar  $r \times s \times 3$  transportation polytope [21], so boundary fibres are not toric in general. Already for  $3 \times 3 \times 3$ , generic instances are simple but not smooth: they are moment polytopes of toric orbifolds [24], not of manifolds [4, §10].*

*Proof.* For  $n_0 = 2$  the second layer is  $m - y$ , where  $y$  is the first layer,  $m$  is the  $(1, 2)$ -margin, and  $y$  has fixed row and column sums with  $0 \leq y \leq m$ . A bipartite incidence matrix stacked with an identity is totally unimodular, so edges are  $\pm 1$  cycle vectors and the argument of theorem 6.4 applies at simple vertices. (ii) combines the cited theorem with the computation in the companion.  $\square$

So the toric regime of the nerve is exactly the 1-simplices and the 2-horns. Boundary fillings leave it, and higher horns can be empty.

## 5 Layer IV: cumulants and the square-zero cone

**Proposition 5.1** (Couplings are second cumulants). *For  $\alpha \in \text{Cpl}(f, g)$ ,  $\alpha - f \otimes g$  is the cross-covariance  $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$  of the indicator vectors, a circulation. The product coupling is the point where all mixed cumulants vanish, and it is the only point of rank one, the Birch point of the independence model [25]. For level-two couplings of second-order states,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \text{Cov}_\Pi(\rho, \rho'),$$

*so flattening keeps only the second mixed cumulant of the two random distributions.*

Fix  $f \in \mathbf{D}(Y)$  and let  $A_f$  be the matrices on  $Y \times Y$  whose row and column sums are multiples of  $f$ . With the gluing product  $a \cdot b = a \operatorname{diag}(f)^{-1} b$  and unit  $\operatorname{diag}(f)$ ,  $A_f$  is an associative unital algebra, spanned by the couplings of  $f$  with itself. The circulations  $I_f$  form a two-sided ideal, and  $A_f/I_f \cong \mathbb{R}$  records the total mass. Whiskering by a kernel  $h$  is a linear unital map  $h_*: A_f \rightarrow A_{h\#f}$ , which need not be multiplicative: a noncommutative expectation in the sense of homotopy probability theory [22, 23]. Its Boolean cumulants [26] are  $b_1 = h_*$ ,  $b_2(a, b) = h_*(a \cdot b) - h_*a \cdot h_*b$ , and so on over interval partitions. The whiskering defect of [theorem 2.3\(ii\)](#) is  $-b_2$ .

**Theorem 5.2** (Whiskering is a two-component  $A_\infty$ -functor). *Let  $C = A_{h\#f} \oplus \varepsilon I_{h\#f}$ , with  $\varepsilon$  of degree  $-1$ ,  $\varepsilon^2 = 0$  and  $d(\varepsilon x) = x$ . Then  $C$  is a differential graded algebra with  $H^0(C) = A_{h\#f}/I_{h\#f} \cong \mathbb{R}$  and  $H^{-1}(C) = 0$ : it is the mapping cone of  $I \hookrightarrow A$ . The maps*

$$F_1 = h_*, \quad F_2 = \varepsilon b_2, \quad F_n = 0 \quad (n \geq 3)$$

*form an  $A_\infty$ -functor  $A_f \rightarrow C$ .*

*Proof.* Every  $b_2$  is a circulation, because both composites couple the same marginals, so  $F_2$  lands in  $\varepsilon I$ .  $C$  is the trivial extension of  $A$  by the bimodule  $\varepsilon I$ , and  $d$  is a derivation because  $I$  is an ideal. Arity two of the functor equation is  $dF_2 = b_2$ , true by definition. In arity three, with  $F_3 = 0$ , it reads  $b_2(ab, c) - b_2(a, bc) = h_*a \cdot b_2(b, c) - b_2(a, b) \cdot h_*c$ . Both sides equal  $h_*a \cdot h_*(bc) - h_*(ab) \cdot h_*c$ , which says that  $b_2$  is the Hochschild coboundary of  $h_*$ . In arity  $n \geq 4$  every term involves some  $F_k$  with  $k \geq 3$  or a product  $F_2 \cdot F_2 \in \varepsilon^2 I = 0$ .  $\square$

*Remark 5.3.* This settles, in the minimal model, the question of the previous draft whether whiskering is an  $A_\infty$ -functor. The higher Boolean cumulants are nonzero [4, §9], but they are not needed here. They appear as higher components only in larger models such as the free cone, as in the bar-construction description of cumulants. The defects span all of  $I$  for three and four points [4, §10], so  $H^0$  of the cone generated by the defects alone is also  $A/I$ .

**The four layers together.** Under the cone, all coupling information is homotopical:  $H^0$  keeps only masses, so the 1-cells of layer I, while the couplings live in degree  $-1$ , in the circulations. These are the tangent spaces of the polytopes of layer III. The cumulants of [proposition 5.1](#), the defects of [theorem 5.2](#) and the deviations of 2-horn fillers from the canonical filler ([theorem 4.2\(iii\)](#)) are all circulations, and the Fisher metric of [proposition 6.5](#) measures them. Layer II carries the tower along the same kernels, with the grades of layer I as bounds.

## 6 Coupling polytopes are moment polytopes

### 6.1 The polytope and its lattice

Let  $f \in \mathbf{D}(Y)$  and  $g \in \mathbf{D}(Y')$  have full support, with  $|Y| = n$ ,  $|Y'| = m$ . (For coupling 2-cells  $Y = Y'$ ; rectangular polytopes arise when gluing contexts [1, Prop. 7.11].) The *coupling polytope* is

$$\operatorname{Cpl}(f, g) = \left\{ \alpha \in \mathbb{R}_{\geq 0}^{Y \times Y'} : \sum_{y'} \alpha(y, y') = f(y), \sum_y \alpha(y, y') = g(y') \right\}.$$

Its tangent space is the space of *circulations*, matrices with zero row and column sums. The integer circulations form a lattice  $\Lambda$  of rank  $(n-1)(m-1)$ ; a basis is  $E_{ij} - E_{i,m} - E_{n,j} + E_{n,m}$  for  $i < n$ ,  $j < m$ .

**Proposition 6.1** (Interior and Fisher metric). *Cpl( $f, g$ ) has dimension  $(n - 1)(m - 1)$ , its relative interior is the set of strictly positive couplings, and on it the Fisher metric  $\mathcal{I}_\alpha(v, w) = \sum v(y, y')w(y, y')/\alpha(y, y')$  is positive definite on circulations.*

*Proof.* The product coupling  $f \otimes g$  is strictly positive, so no inequality  $\alpha \geq 0$  is an implicit equality, and the relative interior of the polytope in its affine hull is  $\{\alpha > 0\}$ . There  $\mathcal{I}_\alpha(v, v) = \sum v^2/\alpha > 0$  for  $v \neq 0$ .  $\square$

*Remark 6.2.* The first draft argued nondegeneracy through a radical and then distinguished “the Fisher metric” from “the Fisher–Rao metric”. These are the same metric, the Fisher information metric. The square-root embedding is a way of computing its geodesic distance [1, Prop. 8.1].

## 6.2 Nondegeneracy implies Delzant

**Definition 6.3** (Walls). The pair  $(f, g)$  is *degenerate* if  $\sum_{y \in I} f(y) = \sum_{y' \in J} g(y')$  for some nonempty proper subsets  $I \subsetneq Y$  and  $J \subsetneq Y'$ . These equations define hyperplanes, the *walls*, in the space of pairs of marginals; the *chambers* are the connected components of the complement of the walls.

**Theorem 6.4** (Coupling polytopes are Delzant). *If  $(f, g)$  is nondegenerate, then Cpl( $f, g$ ) is a Delzant polytope with respect to  $\Lambda$ : it is simple, its facet normals are rational, and at each vertex the primitive edge vectors form a basis of  $\Lambda$ . Its edges at a vertex are the fundamental cycles of a spanning tree.*

*Proof.* Regard  $\alpha$  as a weighting of the edges of the complete bipartite graph on  $Y \sqcup Y'$ . Vertices are forests. If the support of a feasible  $\alpha$  contains a cycle, adding  $\pm\epsilon$  alternately around it stays feasible, so  $\alpha$  is not a vertex. If the support is a forest, the marginal equations determine  $\alpha$  on it uniquely (peel off leaves), so  $\alpha$  is a vertex. *Nondegeneracy makes them spanning trees.* Every node has positive marginal and hence an incident edge of the support. If the support forest had two or more components, take one with row set  $I$  and column set  $J$ . Then  $\sum_I f = \sum_J g$ ;  $I$  and  $J$  are nonempty, and if  $I = Y$  then  $\sum_J g = 1$  forces  $J = Y'$  and the component is everything. So  $I$  and  $J$  are proper, contradicting nondegeneracy. *Simplicity and smoothness.* At a vertex with spanning tree  $T$ , the zero cells are the  $nm - (n + m - 1) = (n - 1)(m - 1)$  non-tree cells  $c$ . Each closes a unique cycle in  $T \cup \{c\}$ , giving a circulation  $C_c$  with entries in  $\{0, \pm 1\}$  and  $C_c(c) = 1$ . A circulation  $w$  equals  $\sum_{c \notin T} w(c)C_c$ , because the difference is a circulation supported on the tree, hence zero. So the feasible directions at the vertex form the cone  $\{\sum_c t_c C_c : t \geq 0\}$  on exactly  $\dim \text{Cpl}(f, g)$  independent generators (simplicity), and the  $C_c$  are integral and form a  $\mathbb{Z}$ -basis of  $\Lambda$ , since integer coefficients  $w(c)$  suffice (smoothness). The facet normals are the coordinate functionals  $\alpha \mapsto \alpha(c)$ , which are rational.  $\square$

By Delzant’s theorem [10] there is a compact connected toric symplectic manifold  $(X_{f,g}, \omega)$  of real dimension  $2(n - 1)(m - 1)$ , unique up to equivariant symplectomorphism, whose moment map  $\mu: X_{f,g} \rightarrow \text{Cpl}(f, g)$  has image the coupling polytope. The fibre over an interior coupling is a Lagrangian torus  $T^{(n-1)(m-1)}$ , and the fibre over a vertex is a point.

**Proposition 6.5** (Fisher metric and Guillemin’s potential). *Suppose every inequality  $\alpha(c) \geq 0$  defines a facet of Cpl( $f, g$ ). The canonical Kähler metric of  $X_{f,g}$  constructed by Guillemin [14] has, in action–angle coordinates over the interior, the form Hess  $G$  in the polytope directions, with symplectic potential*

$$G(\alpha) = \frac{1}{2} \sum_c \alpha(c) \log \alpha(c).$$

Hence the Fisher metric on  $\text{Cpl}(f, g)$  is twice the restriction of the canonical Kähler metric to the real section, and the product coupling  $f \otimes g$ , the coupling of maximal entropy, is the unique critical point of  $G$  on the polytope.

*Proof.* Guillemin's potential is  $\frac{1}{2} \sum_i \ell_i \log \ell_i$  over the facets, where  $\ell_i \geq 0$  are the facet inequalities written with primitive inward normals. Here  $\ell_c(\alpha) = \alpha(c)$ , and the coordinate functional is primitive on  $\Lambda$  because some integer circulation has entry 1 at  $c$ . Then  $\text{Hess } G(v, v) = \frac{1}{2} \sum_c v(c)^2 / \alpha(c)$ . The critical-point condition is that  $\log \alpha$  is orthogonal to all circulations, that is  $\log \alpha(y, y') = r(y) + s(y')$ , which together with the marginals gives  $\alpha = f \otimes g$ . Uniqueness follows from strict convexity.  $\square$

If some cell is not a facet, for instance when  $f(y) + g(y') > 1$  forces  $\alpha(y, y') > 0$ , the potential omits that term and the identity holds only up to those terms.

### 6.3 Walls, chambers and monotone cases

**Proposition 6.6** (Chambers). *The set of spanning trees that support vertices, and hence the combinatorial type of  $\text{Cpl}(f, g)$  and the toric manifold  $X_{f, g}$  up to equivariant diffeomorphism, is constant on each chamber. The type can change across a wall. For  $f = (\frac{1}{2}, \frac{1}{2})$  and  $g = (t, \frac{2}{3} - t, \frac{1}{3})$  the polytope is a hexagon for  $t < \frac{1}{2}$  and a quadrilateral for  $t > \frac{1}{2}$ , both Delzant: two facets become redundant across the wall  $g_1 = f_1$ .*

*Proof.* The tree solution on a spanning tree  $T$  assigns to each tree edge the value  $\sum_I f - \sum_J g$  (up to sign) for the bipartition  $(I, J)$  obtained by deleting that edge. So feasibility of  $T$  depends only on the signs of such differences, which are constant on chambers. The example is checked in `check_conditionals.py`.  $\square$

**Proposition 6.7** (Monotone cases). *Assume  $(f, g)$  is nondegenerate and every cell defines a facet. Then  $X_{f, g}$  is monotone if and only if  $f$  and  $g$  are uniform, and uniform marginals are nondegenerate exactly when  $\gcd(n, m) = 1$ . In particular, in the square case  $Y = Y'$  of coupling 2-cells no nondegenerate pair is monotone. On the wall, the uniform  $2 \times 2$  pair still gives a Delzant segment,  $X = \mathbb{CP}^1$ , with the product coupling as its midpoint; the uniform  $n \times n$  pair with  $n \geq 3$  gives a polytope that is not simple.*

*Proof.* For a Delzant polytope written with primitive normals, the toric manifold is monotone exactly when some interior point has equal lattice distance to all facets: the classes of the toric divisors span  $H^2$  subject to the linear relations given by the normals,  $c_1$  is the sum of the divisor classes, and  $[\omega]$  is the sum weighted by the facet distances. Here the distances are the entries  $\alpha(c)$ , so they are equal exactly when  $\alpha$  is constant, which forces  $f$  and  $g$  to be uniform; then  $\alpha = f \otimes g$ . Uniform marginals are nondegenerate iff  $a/n \neq b/m$  for  $0 < a < n$ ,  $0 < b < m$ , iff  $\gcd(n, m) = 1$ , which fails when  $n = m$ . The two wall cases are checked in `check_conditionals.py` (T3).  $\square$

**Example 6.8** (The degree-6 del Pezzo surface). For  $f$  uniform on two points and  $g$  uniform on three,  $\text{Cpl}(f, g)$  is a hexagon. In the basis of  $\Lambda$  above, its facet normals are  $\pm e_1$ ,  $\pm e_2$  and  $\pm(e_1 + e_2)$ . This is the fan of  $\mathbb{CP}^2$  blown up at three points, the del Pezzo surface of degree 6. All six entries of the product coupling equal  $\frac{1}{6}$ , so it is the equidistant point. The case  $3 \times 4$  (uniform) gives a Delzant polytope of dimension 6 with 96 vertices and all 12 cells facets, again monotone.

## 7 The imported layer: Floer theory of the special fibre

We now quote results from symplectic topology. They are not proved or checked here.

**Nondisplaceability.** For a monotone toric manifold, the torus fibre over the point at equal distance from all facets (the *special fibre*) is a stem in the sense of Entov and Polterovich, hence nondisplaceable [11]. For toric Fano manifolds, Cho and Oh show that a torus fibre with a flat  $U(1)$  connection has nonvanishing Floer cohomology when it corresponds to a critical point of the disc potential [9]; see [8] for the Clifford torus and [12] for general compact toric manifolds.

**Consequence for couplings.** In the monotone cases of [proposition 6.7](#), the special fibre lies over the product coupling  $f \otimes g$ . For [example 6.8](#), the disc potential of the monotone special fibre is, up to the factor  $T^{1/6}$ , the Laurent polynomial

$$W(x, y) = x + y + \frac{1}{x} + \frac{1}{y} + xy + \frac{1}{xy},$$

read off from the facet normals, and it has a critical point at  $(x, y) = (1, 1)$  (checked exactly), so the special fibre with trivial holonomy has nonzero Floer cohomology by [9].

The maximal coupling, by contrast, has zero entries whenever  $f \neq g$ , so it lies on the boundary of  $\text{Cpl}(f, g)$ , where the torus fibre collapses (`check_conditionals.py`, T6). The first draft’s conjecture that the critical points are the maximal couplings should therefore be replaced. The geometrically distinguished coupling is the product coupling, which is also the maximum-entropy coupling and the critical point of Guillemin’s potential ([proposition 6.5](#)).

## 8 Stability conditions: a question

This subsection replaces a draft that proposed a Bridgeland stability condition whose central charge is the disc potential, whose heart is the committed states, and whose Harder–Narasimhan filtration is the obstruction tower. None of those identifications is well typed. A central charge is an additive map  $K_0 \rightarrow \mathbb{C}$  and  $W$  is a function on a torus. Committed states are vertices of a polytope and form no abelian category. The obstruction tower assigns numbers to a claim; it does not filter an object. What can be said precisely is the following.

**Bridgeland stability.** A stability condition on a triangulated category  $\mathcal{D}$  is a pair  $(Z, \mathcal{A})$ . Here  $\mathcal{A}$  is the heart of a bounded  $t$ -structure, and  $Z: K_0(\mathcal{D}) \rightarrow \mathbb{C}$  is additive, sends nonzero objects of  $\mathcal{A}$  to the upper half-plane together with  $\mathbb{R}_{<0}$ , and admits Harder–Narasimhan filtrations; the support property is also required [6]. On any smooth projective surface  $X$ , and for any ample class  $\omega$ , there is a stability condition with central charge

$$Z_\omega(E) = - \int_X e^{-i\omega} \text{ch}(E)$$

whose heart is a tilt of  $\text{Coh } X$  [3, 16]. (A  $B$ -field can be added; we set it to zero.)

**Proposition 8.1** (Marginals give stability conditions on the  $2 \times 3$  surfaces). *Let  $(f, g)$  be nondegenerate in the  $2 \times 3$  family, let  $X_\Sigma$  be the toric surface of its chamber, and let  $D_c$  be the toric divisor of the facet  $\{\alpha_c = 0\}$ . Put*

$$[\omega_{f,g}] = \sum_{c \text{ facet}} f_i g_j [D_c], \quad c = (i, j),$$

*the product coupling read as divisor coefficients. Then:*

- (i)  $[\omega_{f,g}]$  is the class of the toric symplectic form with moment polygon  $\text{Cpl}(f, g)$ . It is independent of the base point used to write the polygon, and it is ample.
- (ii)  $\omega_{f,g} \cdot D_c$  is the lattice length of the edge  $\{\alpha_c = 0\}$  of the coupling polygon, and  $\frac{1}{2}\omega_{f,g}^2$  is its lattice area.
- (iii) There is a stability condition  $\sigma_{f,g} = (Z_{\omega_{f,g}}, \mathcal{A}_{\omega_{f,g}})$  on  $D^b(\text{Coh } X_\Sigma)$ . It depends continuously on  $(f, g)$  within the chamber, and

$$Z(\mathcal{O}_X) = \text{area } \text{Cpl}(f, g), \quad Z(\mathcal{O}_x) = -1, \quad Z(\mathcal{O}_{D_c}) = \frac{1}{2}D_c^2 + i \text{ length}\{\alpha_c = 0\}.$$

- (iv) As  $(f, g)$  approaches a wall from inside a chamber, one edge length tends to 0, its divisor  $E$  is a  $(-1)$ -curve, and  $Z(\mathcal{O}_E) \rightarrow -\frac{1}{2}$ , a point on the negative real axis.

*Proof.* (i) If a Delzant polygon is  $\{u : \langle u, v_c \rangle + \lambda_c \geq 0\}$ , the symplectic class is  $\sum_c \lambda_c [D_c]$ , up to the normalization of  $\omega$ . Here the facet functions are  $\alpha_c(u) = \langle u - u_0, v_c \rangle + \alpha_c(u_0)$  for any interior base point  $u_0$ , and choosing  $u_0 = f \otimes g$  gives the formula. Another choice of base point changes the coefficients by  $\langle w, v_c \rangle$ , which is a linear relation among the  $[D_c]$ . A Kähler class on a projective toric surface is ample. (ii) These are standard toric identities: the degree of  $\omega$  on  $D_c$  is the lattice length of the corresponding edge, and the symplectic volume is the lattice area (Duistermaat–Heckman). (iii) Apply [3, 16] with  $\omega = \omega_{f,g}$ , which varies continuously with  $(f, g)$ . The values follow from  $\text{ch}(\mathcal{O}_X) = 1$ ,  $\text{ch}(\mathcal{O}_x) = [\text{pt}]$  and  $\text{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$ . (iv) By [proposition 6.6](#) and `check_stability_layer.py` (S5), the vanishing edge is that of a  $(-1)$ -curve. All identities are checked in all 18 chambers.  $\square$

So the probabilistic data enter Bridgeland stability through the Kähler class, and nowhere else. The product coupling fixes the class, edge lengths of the coupling polygon are imaginary parts of central charges, and its area is  $Z(\mathcal{O}_X)$ . At the uniform point  $\omega = -\frac{1}{6}K_X$  on the del Pezzo surface, with area  $\frac{1}{12}$  and six edges of length  $\frac{1}{6}$ .

## Questions.

- (Q1) *Across walls.* Crossing a wall blows up or blows down a point ([proposition 6.6](#)), and Orlov’s theorem gives  $D^b(\text{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X) \rangle$  [17]. As the wall is approached,  $Z(\mathcal{O}_E)$  tends to the negative real axis. Does the path  $\sigma_{f,g}$  extend across the wall to a path through stability conditions glued along Orlov’s decomposition? If so, the chamber structure of the marginals becomes a wall structure in a single space of stability conditions.
- (Q2) *Objects for couplings.* Is there a natural object  $E_\alpha \in D^b(\text{Coh } X)$  attached to an interior coupling  $\alpha$ ? A candidate is the SYZ or mirror transform of the torus fibre over  $\alpha$  with a flat connection. Does  $Z(E_\alpha)$  have a probabilistic meaning, for instance in terms of the Fisher metric of [proposition 6.5](#)?
- (Q3) *Quantum corrections.* Iritani’s integral structure identifies quantum-corrected central charges of objects of  $D^b(\text{Coh } X)$  with oscillatory integrals  $\int e^{-W/z}$  of the mirror potential over integral cycles, and their asymptotics involve the  $\widehat{\Gamma}$ -class [15, 13]. This is the precise sense in which the disc potential is related to central charges: through oscillatory integrals, not through its values. Do the corrected central charges, with  $\omega = \omega_{f,g}$ , define stability conditions on these surfaces? Does the monotone point  $f \otimes g$  uniform play a special role?

(Q4) *Higher dimensions.* For  $(n-1)(m-1) \geq 3$  the toric manifolds have complex dimension at least three. There, the existence of stability conditions is known only in special cases, through Bogomolov–Gieseker type inequalities [5]. The construction of [proposition 8.1](#) therefore extends only conditionally.

**What is not claimed.** We make no identification of the disc potential with a central charge, of committed states with a heart, of the obstruction tower with Harder–Narasimhan filtrations, or of hallucination with instability. The obstruction tower and the stability conditions above live on different objects (claims and admissibility data on one side, coherent sheaves on  $X_{f,g}$  on the other), and any relation between them would have to be constructed. Hallucination keeps a precise meaning, but on the tower side: a claim whose required coherent lift is obstructed [2, Def. 10.17], a convex obstruction with a certificate. Unfillable horns are instances of it ([theorem 11.2](#)). [Table 1](#) lists each withdrawn identification, the reason, and what is claimed instead.

## 9 The conjectural layer

**Proposition 9.1** (The  $2 \times 3$  family [4, §3]). *In the  $2 \times 3$  family there are 18 chambers. Every toric surface  $X_{f,g}$  is the degree-6 del Pezzo surface or a toric blow-down of it, and crossing one wall is exactly one toric blow-up or blow-down.*

**Conjecture 9.2** (Family over the chamber complex). *For general  $n \times m$ , the toric manifolds  $X_{f,g}$  form a family over each chamber, and crossing a wall is a toric birational transformation (in higher dimension possibly a flip) determined by the pair  $(I, J)$  defining the wall. On a wall,  $X_{f,g}$  degenerates to a singular toric variety whose moment polytope is the non-simple  $\text{Cpl}(f, g)$ .*

**Question 9.3** (Location of Floer-nontrivial fibres). Outside the monotone cases, the fibres with nonvanishing Floer cohomology (possibly after bulk deformation) are located by the potential of [12]. Is the product coupling always among them, and do they have a statistical meaning?

**Question 9.4** (Beyond the minimal cone). [Theorem 5.2](#) makes whiskering an  $A_\infty$ -functor into the square-zero cone, with two components. (i) In the free cone, do the higher Boolean cumulants give the higher components? (ii) Does the construction extend to the multi-object setting, an  $A_\infty$ -functor between linear categories whose objects are 1-cells, compatible with the nerve of [section 4](#)?

**Conjecture 9.5** (Fukaya-type functor). *There is a functor from a category built from the Fukaya categories of the  $X_{f,g}$  (with special fibres as distinguished objects) to a linearization of the coupling 2-cells, compatible with gluing on one side and composition of Lagrangian correspondences on the other.*

## 10 Experimental definitions and evidence

This section takes up the five problems left open by the previous draft. Each subsection states a definition, proves what can be proved, and records what was only computed. The numbers are tabulated in [4, §13], and the scripts are listed in [section 13](#).

### 10.1 The multi-object cone

**Definition 10.1** (The linear category of couplings). Let  $\mathcal{L}$  be the category whose objects are full-support distributions  $f$  on finite sets. Its morphisms are

$$\mathcal{L}(f, g) = V(f, g) = \{a \in \mathbb{R}^{Y \times Y'} : a\mathbf{1} = s f, \mathbf{1}^\top a = s g^\top \text{ for some } s = s(a) \in \mathbb{R}\},$$

with composition  $a \cdot b = a \operatorname{diag}(g)^{-1}b$  and identity  $1_f = \operatorname{diag}(f)$ . Let  $\mathcal{I}(f, g) = \{a : s(a) = 0\}$  be the circulations, and let  $\mathcal{C}$  be the mapping cone of  $\mathcal{I} \hookrightarrow \mathcal{L}$ : it has  $\mathcal{C}(f, g) = V(f, g) \oplus \varepsilon \mathcal{I}(f, g)$ , with  $\deg \varepsilon = -1$ ,  $m_1(\varepsilon x) = x$  and  $\varepsilon x \cdot \varepsilon y = 0$ . For a family of kernels  $h = (h_Y : Y \rightarrow Z_Y)$ , put  $F_1^h(a) = h_Y^\top a h_{Y'}$ ,  $F_2^h = \varepsilon \delta_h$  with  $\delta_h(a, b) = F_1^h(a \cdot b) - F_1^h(a) \cdot F_1^h(b)$ , and  $F_n^h = 0$  for  $n \geq 3$ .

Couplings are the morphisms with  $a \geq 0$  and  $s = 1$ , and their composition is gluing. When  $f = g$  this is the algebra  $A_f$  of [section 5](#). For kernels  $X \rightarrow Y$  as objects, take the product over  $x \in X$ .

**Theorem 10.2** (Multi-object version of [theorem 5.2](#)). *(i)  $\mathcal{L}$  is a linear category,  $s$  is a functor to the one-object category  $\mathbb{R}$ , and its kernel  $\mathcal{I}$  is a two-sided ideal with  $\dim \mathcal{I}(f, g) = (|Y| - 1)(|Y'| - 1)$ . Hence  $H^0(\mathcal{C}) = \mathcal{L}/\mathcal{I}$  has every hom-space equal to  $\mathbb{R}$ , and  $H^{-1}(\mathcal{C}) = 0$ .*

*(ii)  $F^h$  is an  $A_\infty$ -functor  $\mathcal{L} \rightarrow \mathcal{C}$  with two components. It is cohomologically unital but not strictly unital:  $F_1^h(1_f) - 1_{h\#f} = m_1(\varepsilon u_f)$  with  $u_f \in \mathcal{I}$ , and  $F_2^h(1_f, a) \neq 0$  in general.*

*(iii) Whiskering is strictly functorial in the kernel:  $F^{h'} \circ F^h = F^{hh'}$  as  $A_\infty$ -functors, where  $F^{h'}$  acts on  $\varepsilon \mathcal{I}$  by  $F_1^{h'}$ .*

*Proof.* (i)  $(a \operatorname{diag}(g)^{-1}b)\mathbf{1} = s(b)a\mathbf{1} = s(a)s(b)f$ , and similarly for column sums, so composition is well typed and  $s$  is multiplicative. The dimension count is that of the transportation polytope's tangent space. (ii)  $F_1^h$  preserves  $s$  because kernels preserve mass, so  $\delta_h$  and  $u_f$  have  $s = 0$ . The arity-two and arity-three equations are those in the proof of [theorem 5.2](#), which never used that source and target coincide. (iii)  $F_1$  composes because  $(hh')^\top a (hh') = h'^\top (h^\top a h) h'$ . The composite's second component is  $F_1^{h'} \delta_h(a, b) + \delta_{h'}(F_1^h a, F_1^h b)$ , which telescopes to  $\delta_{hh'}(a, b)$ .  $\square$

**Definition 10.3** (Nerve to cone). For a 2-simplex  $p$  of the coupling nerve, with edges  $d_2 p \in V(f_0, f_1)$ ,  $d_0 p \in V(f_1, f_2)$  and  $d_1 p \in V(f_0, f_2)$ , put  $H(p) = \varepsilon(d_1 p - d_2 p \cdot d_0 p) \in \varepsilon \mathcal{I}(f_0, f_2)$ .

**Proposition 10.4.**  $m_1 H(p) = d_1 p - d_2 p \cdot d_0 p$ , so every 2-simplex is a homotopy from the composite of two of its edges to the third.  $H$  vanishes on gluing fillers and on degenerate 2-simplices. It is natural for whiskering:  $H(h_* p) = F_1^h(H(p)) + F_2^h(d_2 p, d_0 p)$ . On a 3-simplex the coherence class  $Z = \gamma_{01} H_{123} + H_{013} - H_{012} \gamma_{23} - H_{023}$  vanishes identically.

*Proof.* The first two claims follow from the definition and [theorem 4.2\(iii\)](#). Naturality is the identity  $h_*(d_1 p) - h_* d_2 p \cdot h_* d_0 p = h_*(d_1 p - d_2 p \cdot d_0 p) + \delta_h(d_2 p, d_0 p)$ . Expanding  $d_1$  of each face of a 3-simplex gives  $m_1 Z = 0$ , and  $m_1$  is injective on  $\varepsilon \mathcal{I}$ .  $\square$

So [question 9.4\(ii\)](#) has a positive answer on the 2-skeleton. The answer is also limited: since  $H^{-1}(\mathcal{C}) = 0$ , the square-zero cone cannot see the 3-simplex obstructions of [theorem 4.3](#). Detecting contextuality needs a target with nonzero  $H^{-1}$ , such as the free cone of [question 9.4\(i\)](#).

## 10.2 Wall-crossing as variation of GIT

**Proposition 10.5** (Cox presentation and walls).  $X_{f,g}$  is the GIT quotient  $\mathbb{C}^{n \times m} //_{(f,g)} (T^n \times T^m) / \Delta$ , where the cell  $(i, j)$  has weight  $e_i \oplus e_j$ . The kernel of the ray map  $\mathbb{Z}^{n \times m} \rightarrow \Lambda^*$ ,  $c \mapsto v_c$ , is the saturated lattice of matrices  $r_i + s_j$ , of rank  $n + m - 1$ . The hyperplanes in the character space  $\{\sum f = \sum g\}$  spanned by weights are exactly the walls  $\sum_I f = \sum_J g$  and the facets  $f_i = 0$ ,  $g_j = 0$  of the weight cone.

*Proof.* The first statement is Cox's construction [28] for the fan of [section 6](#), with the kernel checked in `check_vgit.py` (V1). For the second, a functional  $(a, b)$  vanishes on the weights of a set  $S$  of cells if and only if  $a_i + b_j = 0$  for  $(i, j) \in S$ , that is, if and only if  $a$  and  $-b$  are constant on the connected components of the bipartite graph  $S$ . The rank of  $S$  is  $n + m$  minus the number of components. It is  $n + m - 2$  exactly when there are two components,  $I \sqcup J$  and  $I^c \sqcup J^c$ . Then the functional is  $\sum_I f - \sum_J g$ , or  $f_i$  or  $g_j$  when a component is a single isolated vertex.  $\square$

So [conjecture 9.2](#) is an instance of variation of GIT for torus actions [30, 29]. The quotient is constant on chambers and changes by a birational transformation at each wall. What remains is the type of transformation. In the examples, every single-wall crossing is one toric blow-up or blow-down, or a flip with the same rays. For  $2 \times 4$ , 40 random crossings gave 18 blow-ups or blow-downs and 22 flips. For  $3 \times 3$ , 25 crossings gave 6 and 19 (`check_vgit.py` (V3)).

### 10.3 Across a wall

Follow the path  $P(t)$  of [4, §12] from the pentagon chamber ( $X_- = \text{Bl}_2 \mathbb{CP}^2$ ) to the hexagon chamber ( $X_+$  the del Pezzo surface of degree 6). It crosses only the wall  $f_1 = g_1$ , at  $t = \frac{3}{8}$ . Let  $\pi: X_+ \rightarrow X_-$  blow up the fixed point  $D_{01} \cap D_{02}$ , with exceptional curve  $E = D_{10}$ , and let  $s(t) = f_1 - g_1$ , the lattice length of the new edge.

**Proposition 10.6** (Central charges across the wall). *Let  $\omega_-(t)$  be the pentagon class with its support numbers continued affinely past the wall. Then:*

- (i)  $\omega_+ = \pi^* \omega_- - sE$ , so  $\text{area}_+ = \text{area}_- - \frac{1}{2}s^2$ . In particular the jump  $\frac{16}{225}$  of the second derivative of the area [4, §12] is  $(ds/dt)^2$ .
- (ii)  $Z_+(\pi^* F) = Z_-(F) - \text{rk}(F) \frac{1}{2}s^2$  for all  $F \in K_0(X_-)$ , and  $Z_+(\mathcal{O}_E(-1)) = \frac{1}{2} + is$ .
- (iii) At the level of  $K_0$ , Orlov's decomposition  $\langle \mathcal{O}_E(-1), \pi^* D^b(X_-) \rangle$  holds:  $\chi(\mathcal{O}_E(-1), \mathcal{O}_E(-1)) = 1$ ,  $\chi(\pi^* F, \mathcal{O}_E(-1)) = 0$ , and  $K_0(X_+) = \mathbb{Z}[\mathcal{O}_E(-1)] \oplus \pi^* K_0(X_-)$ .

*Proof.* (i)  $\pi^*$  is an isometry onto  $E^\perp$  with  $E^2 = -1$ , and the difference of support numbers on  $E$  is  $h_{10} - h_{01} - h_{02} = g_1 - f_1$ . (ii)  $\text{ch}$  commutes with  $\pi^*$ . Then  $(\pi^* \omega - sE) \cdot \pi^* D = \omega \cdot D$  and  $(\pi^* \omega - sE)^2 = \omega^2 - s^2$ , and  $\text{ch } \mathcal{O}_E(-1) = (0, E, -\frac{1}{2})$ . (iii) This is Riemann–Roch, checked on generators of  $K_0(X_-)$ . All of it is verified exactly along the path in `check_across_wall.py`.  $\square$

**Definition 10.7** (Glued central charge). Along the path, define  $Z_t$  on  $K_0(X_+) = \mathbb{Z}[\mathcal{O}_E(-1)] \oplus \pi^* K_0(X_-)$  by  $Z_t = Z_-$  on  $\pi^* K_0$  for  $t \leq \frac{3}{8}$ , and by  $Z_+$  for  $t > \frac{3}{8}$ , with  $Z_t(\mathcal{O}_E(-1)) = \frac{1}{2} + i s(t)^+$ .

By [proposition 10.6](#),  $Z_t$  is continuous at the wall, and the new class enters at the real point  $\frac{1}{2}$ : the phase of  $\mathcal{O}_E(-1)$  tends to 0, and that of  $\mathcal{O}_E$  tends to 1. This answers (Q1) of [section 8](#) at the level of central charges. The glued charges extend continuously, in the form the gluing construction of [27] requires along Orlov's decomposition. The hearts and the gluing condition on them are not tested.

### 10.4 The Fukaya probe

**Definition 10.8** (Candidate object map). For  $u$  in the interior of  $\text{Cpl}(f, g)$ , let  $\mathfrak{PD}(u; w) = \sum_c \exp(-\lambda \alpha_c(u) + \langle n_c, w \rangle)$ , with  $T = e^{-\lambda}$ . A critical point  $w$  whose real part stays bounded as  $\lambda \rightarrow \infty$  makes the fibre  $L(u)$  Floer-nontrivial, with holonomy  $\text{Im } w$  [12]. The candidate object map of [conjecture 9.5](#) sends  $(L(u), \text{Im } w)$  to the coupling  $u$ .

Since  $\mathfrak{P}\mathfrak{D}(u; w) = \mathfrak{P}\mathfrak{D}(0; w - \lambda u)$ , the locations  $u_k = -\lim \operatorname{Re} w_k / \lambda$  of the critical points of  $\mathfrak{P}\mathfrak{D}(0; \cdot)$  are intrinsic.

**Proposition 10.9** (Monotone point). *At  $f = (\frac{1}{2}, \frac{1}{2})$ ,  $g = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ , the six critical points of  $W$  (example 6.8) all lie over the product coupling. The Hessian of  $W$  in logarithmic coordinates is a positive multiple ( $\frac{1}{6}$ ) of the Fisher metric  $\sum_c n_c n_c^\top / \alpha_c$  at the product coupling at the critical point  $(1, 1)$ , and at no other.*

*Proof.* All  $\alpha_c$  equal  $\frac{1}{6}$  at the product coupling, so  $\operatorname{Hess} W(1, 1) = \sum_c n_c n_c^\top = \frac{1}{6} \cdot \text{Fisher}$ . At the other critical points the coefficients  $x^{n_c}$  are non-real or have mixed signs (`check_fukaya_probe.py`, K3).  $\square$

**Observation 10.10** (Off the monotone point). At  $(f_1, g_1, g_2) = (0.52, 0.31, 0.35)$  in the hexagon chamber, the six critical points, tracked to  $\lambda = 2560$ , lie over three couplings. Four lie over the *leximin coupling*, which maximizes the smallest cell, then the next smallest, and so on. One lies over each of two couplings where three minimal cells balance with holonomy signs,  $\sum_c \pm n_c = 0$ . The product coupling is not among them. Off the monotone point the Hessian of  $\mathfrak{P}\mathfrak{D}$  carries Boltzmann weights  $e^{-\lambda \alpha_c}$ , not the Fisher weights  $1/\alpha_c$ , and the two forms are not proportional. In `fukaya_scan.py`, which finds all critical points by elimination at  $\lambda = 500$  and  $2000$ , the same pattern holds at three further random points of the chamber. There are three locations with multiplicities  $4 + 1 + 1$ , and the leximin coupling carries four of them, including the positive real one.

This answers [question 9.3](#) negatively for the product coupling. The Floer-nontrivial fibres do have a statistical meaning: the leximin coupling, the most balanced coupling in the max–min sense, consistent with the iterated max–min construction of a nondisplaceable fibre in [12], together with signed-balanced couplings. The candidate object map of [definition 10.8](#) therefore lands on these couplings, not on the maximum-entropy coupling. The metric compatibility hoped for in [conjecture 9.5](#) holds only at the monotone point.

## 11 The obstruction tower and the nerve

The tower paper [2] attaches obstructions to a claim. Given admissibility data  $S_1 \subseteq DB$  and  $S_2 \subseteq D(S_1)$ , it uses three kinds of set: local sets  $L_1 = \operatorname{conv} S_1$  and  $L_2 = D(S_1)$ , effective sets  $E_2 = \operatorname{conv} S_2$  and  $E_1 = \mu_B(E_2)$ , and the fibre  $\mathcal{F}(x_1) = \{\Pi \in L_2 : \mu_B \Pi = x_1\}$  of level-two lifts. The *primary obstruction* is the failure of the local test  $x_1 \in L_1$ . The *secondary obstruction* is the interval  $I(x_1) = o_3(\mathcal{F}(x_1)) = [o_3^-, o_3^+]$ , where  $o_3(\Pi)$  is the  $\ell^1$  distance from  $\Pi$  to  $E_2$  [2, Def. 10.11, Def. 10.14, Prop. 10.15]. This section states how these obstructions relate to layers III and IV and to the toric geometry. It also records which identifications fail.

**Typing at level two.** In a coherent context  $(\rho, \Pi, \Omega, \Sigma)$  the first component is determined by the second:  $\rho = \mu_B \Pi$ . The pair  $(\rho, \Pi)$  is therefore a second-order state, not a coupling. Couplings enter between two contexts. For kernels  $f, g$  and a context  $c$ , the level-two grade of [proposition 3.4](#) is a Kantorovich distance, that is, a minimum over the level-two coupling polytope  $\operatorname{Cpl}(A_f(c)_2, A_g(c)_2)$  of couplings of two second-order states. This polytope has dimension  $(|\operatorname{supp} A_f(c)_2| - 1)(|\operatorname{supp} A_g(c)_2| - 1)$  and is Delzant when nondegenerate [4, Prop. 5.1]. Flattening maps it affinely to  $\operatorname{Cpl}(\mu A_f(c)_2, \mu A_g(c)_2)$ , and its fibres have fixed cross-covariance ([proposition 5.1](#)). So the toric geometry acts on pairs of transported contexts, not on a single context.

**Definition 11.1** (A horn as a tower). Let  $\Lambda$  be a horn on finite sets with faces  $F_1, \dots, F_n$ . Put  $B = \bigsqcup_i Y^{F_i}$ . Encode horn data  $(p_i)_i$  as the level-one claim  $x_1 = \frac{1}{n} \sum_i p_i$  in  $DB$ , and a global assignment  $s$  as  $\rho_s = \frac{1}{n} \sum_i \delta_{s|F_i}$ . Let  $\text{Loc} \subseteq DB$  be the encoded compatible families, the local polytope of [19]. Take  $S_1 \subseteq \text{Loc}$  finite and containing every  $\rho_s$ , and  $S_2 = \{\delta_{\rho_s}\}$ : only global mechanisms are effective.

**Theorem 11.2** (Horn obstructions are secondary obstructions). *In definition 11.1:*

- (i) If  $S_1$  is the vertex set of  $\text{Loc}$ , then  $L_1 = \text{Loc}$ , so the primary test is compatibility of the horn data.
- (ii)  $E_1 = \text{conv}\{\rho_s\}$ , and  $x_1 \in E_1$  if and only if the horn fills. An unfillable horn is an extension-obstructed claim in the sense of [2, Def. 10.17]: it passes the local test but has no coherent lift.
- (iii)  $o_3^-(x_1) \geq 2 \text{CF}(x_1)$ , with equality when  $S_1$  is the vertex set of  $\text{Loc}$ . Here  $\text{CF}$  is the contextual fraction of theorem 4.3(iii). The inequality can be strict when  $S_1$  is smaller: if  $S_1$  consists of the  $\rho_s$  and the parity families, then  $o_3^- = 2$  on the parity  $n$ -horn, while  $2 \text{CF} = 2 - 2^{3-n}$ .
- (iv)  $d(x_1, E_1) \leq 2 \text{CF}(x_1)$  in  $\ell^1$ .
- (v) If  $S_1 = \{\rho_s\}$ , then  $\mathcal{F}(x_1)$  is the polytope of fillers. For a 2-horn its directions are the slice-wise circulations of theorem 4.2(iii).

*Proof.* (i) Deterministic compatible families are exactly the  $\rho_s$ , and  $\text{Loc}$  is the convex hull of its vertices. (ii)  $\mu_B(S_2) = \{\rho_s\}$ . A global law  $G$  restricts to the data exactly when  $\sum_s G(s)\rho_s = x_1$ , because the faces sit in disjoint summands of  $B$ . (iii)  $E_2 = D(\{\rho_s\})$  inside  $D(S_1)$ , so  $o_3(\Pi) = 2\Pi(S_1 \setminus \{\rho_s\})$ , as in [2, Prop. 10.9]. Minimizing over  $\mathcal{F}(x_1)$  gives twice the least weight on non-global mechanisms over all decompositions of  $x_1$  within  $S_1$ . This is at least the least  $w$  with  $x_1 = (1-w)e + we'$ ,  $e \in \text{conv}\{\rho_s\}$ ,  $e' \in \text{Loc}$ . When  $S_1$  contains all vertices of  $\text{Loc}$ , every such  $e'$  decomposes within  $S_1$ , so equality holds. The least  $w$  is  $\text{CF}$ : if  $\sum_s \lambda_s \rho_s \leq x_1$ , the remainder is non-negative and satisfies the linear compatibility equations, so the remainder constraint is automatic [18]. For the parity horns no decomposition within the smaller  $S_1$  uses a global mechanism, which `check_tower_horns.py` confirms for  $n = 3, 4, 5$ . (iv) This is [2, Prop. 5.12], since  $\text{CF} = \mu_{\min}$  with the  $\rho_s$  as sources. (v)  $\mu_B \Pi = x_1$  says that the global law  $\Pi$  restricts to the data, and then apply theorem 4.2.  $\square$

**Corollary 11.3** (Tower levels are not horn dimensions). *Every horn obstruction, of any dimension  $n \geq 3$ , appears at the same place in the tower: as an extension-obstructed level-one claim measured by the secondary obstruction. The primary test detects incompatible face data, not the failure of a 2-horn, since 2-horns always fill. Level three records uncertainty about the decomposition itself, a law on second-order states. The tower levels count orders of uncertainty, while the horn dimension counts the arity of a joint law. So the primary, secondary and tertiary obstructions do not correspond to 2-, 3- and 4-horns.*

For the parity horns,  $d(x_1, E_1) = \frac{1}{3}, \frac{3}{8}, \frac{9}{20}$  for  $n = 3, 4, 5$ , which is twice the best approximate filler distance  $\varepsilon^*(n)$  of [4, §13.1]. One inequality is general: since  $x_1$  averages the face laws,  $d(x_1, E_1) \leq 2\varepsilon^*$ . Equality is computed, not proved.

**Proposition 11.4** (Information bounds on the obstructions). *Let  $y \in E_1$  have full support. Then*

$$d(x_1, E_1)^2 \leq \|x_1 - y\|_1^2 \leq \min\{\chi^2(x_1\|y), 2 \text{KL}(x_1\|y)\}, \quad \chi^2(x_1\|y) = g_y^F(x_1 - y, x_1 - y),$$

where  $g^F$  is the Fisher metric and  $\text{KL}$  is the Bregman divergence of  $\Phi(y) = \sum_b y(b) \log y(b)$ , with  $\text{Hess } \Phi = g^F$ . The same bounds hold for  $o_3(\Pi)$  with  $y$  replaced by any  $\Pi' \in E_2$  of full support. The Fisher bound is an equality exactly when  $|x_1 - y|$  is proportional to  $y$ . No reverse inequality holds, since  $\text{KL}$  is infinite when  $y$  misses part of the support of  $x_1$ .

*Proof.* By Cauchy–Schwarz,  $(\sum |v|)^2 = (\sum (|v|/\sqrt{y})\sqrt{y})^2 \leq \sum v^2/y$ , with equality exactly when  $|v|/\sqrt{y}$  is proportional to  $\sqrt{y}$ . Pinsker’s inequality gives the  $\text{KL}$  bound.  $\square$

On a coupling polytope,  $\Phi = \sum_c \alpha_c \log \alpha_c$  is twice the Guillemin potential of [proposition 6.5](#). So obstruction distances between couplings are bounded by its Bregman divergence and by the Fisher form, and the circulations are the directions in which these are measured. In the tower example of [\[4, §11\]](#), with  $x_1 = (\frac{7}{10}, \frac{3}{10})$  and  $E_1 = \{(\frac{1}{2}, \frac{1}{2})\}$ , the Fisher bound is attained:  $\|x_1 - y\|_1^2 = \chi^2 = \frac{4}{25}$ .

**Proposition 11.5** (Transport can only lower the secondary obstruction). *For a kernel  $k: B \rightarrow B'$ , transport the data to  $k_{\#}S_1$  and  $S_2$  pushed forward. Then  $o_3^-(k_{\#}x_1) \leq o_3^-(x_1)$  and  $d(k_{\#}x_1, E'_1) \leq d(x_1, E_1)$ .*

*Proof.* Pushforward along a kernel is an  $\ell^1$  contraction, pushes  $\mathcal{F}(x_1)$  into  $\mathcal{F}(k_{\#}x_1)$  and maps  $E_k$  onto  $E'_k$ . Take the minimizers before transport, push them forward, and compare.  $\square$

**What is not identified.** The obstructions are numbers, values of linear programs, and their certificates are test functions on  $B$ , on  $S_1$  or on assignments. Circulations are not obstructions. They are the directions of indeterminacy: tangent vectors of fibres and of filler polytopes, measured by the Fisher metric. Boolean cumulants are multilinear maps on composable couplings, while the tower’s obstructions are attached to a single claim and indexed by order of uncertainty. No map sends one to the other, and the cumulant order is not the tower level. The monotonicity of failure levels [\[2, Thm. 10.5\]](#) and [proposition 11.5](#) resemble properties of Harder–Narasimhan filtrations, but no filtration of an object is involved, and none is claimed.

## 12 Next steps

The companion [\[4\]](#) computes every layer explicitly. Of the five problems left open by the previous draft, [section 10](#) settles or reduces each:

- (1) *Across a wall:* the central charges glue continuously ([proposition 10.6](#)). Open: the hearts, and the gluing condition of [\[27\]](#) on them.
- (2) *General wall-crossing:* [conjecture 9.2](#) is toric VGIT ([proposition 10.5](#)). Open: prove that a single wall is always a blow-up, a blow-down or a flip, as in all computed  $2 \times 4$  and  $3 \times 3$  crossings, and read off which from  $(I, J)$ .
- (3) *Higher horns:* the parity horns have contextual fraction  $1 - 2^{2-n}$  for every  $n$  ([theorem 4.3](#)). Open: a closed form for the best approximate fillers  $\varepsilon^*(n)$ , which increase and stay below  $\frac{1}{2}$  for  $n \leq 8$ .
- (4) *Multi-object cone:* positive on the 2-skeleton ([theorem 10.2](#) and [proposition 10.4](#)). Open: a target with  $H^{-1} \neq 0$  that sees 3-simplex obstructions, which is [question 9.4\(i\)](#).
- (5) *The tower:* horn obstructions are secondary obstructions of encoded claims ([theorem 11.2](#)). Open: a proof that  $d(x_1, E_1) = 2\varepsilon^*$  for symmetric horns, and an encoding of level three.

- (6) *Fukaya*: objects go to leximin and signed-balanced couplings, and the Fisher metric matches only at the monotone point ([proposition 10.9](#), [observation 10.10](#)). Open: a proof that leximin always carries the trivial-holonomy critical point, and the functor on morphisms.

## 13 Computational checks

All checks pass.

- `check_glue.py` (30 checks): [proposition 2.1](#) and [theorem 2.3](#).
- `check_stochastic_basechange.py` (13 checks): [theorem 3.1](#); the hexagon of [example 6.8](#); the Delzant property for random nondegenerate  $3 \times 3$  marginals; non-simplicity of the uniform  $3 \times 3$  case; the Hessian identity and critical point of [proposition 6.5](#).
- `check_conditionals.py` (7 checks): [theorem 6.4](#), [propositions 6.6](#) and [6.7](#), and [example 6.8](#).
- `check_stability_layer.py` (6 checks): [proposition 8.1](#) in all 18 chambers of the  $2 \times 3$  family.
- In the companion [\[4\]](#):
  - `check_tower_fibration.py` (7 checks): [theorem 3.1](#) and [proposition 3.4](#);
  - `check_nerve.py` (10 checks): [theorems 4.2](#), [4.3](#) and [5.2](#) and [proposition 4.5](#);
  - `check_cumulants.py` (8 checks): [proposition 5.1](#);
  - `check_polytope_companion.py`, `check_family.py` and `check_family_geometry.py`: [proposition 9.1](#).
- For [section 10](#):
  - `check_horns_n.py` (4 checks): [theorem 4.3\(iii\)](#) for all  $n$ , and  $\varepsilon^*(n)$  for  $n \leq 8$ ;
  - `check_vgit.py` (4 checks): [proposition 10.5](#) and the crossing types;
  - `check_multiobject_cone.py` (9 checks): [theorem 10.2](#) and [proposition 10.4](#);
  - `check_across_wall.py` (7 checks): [proposition 10.6](#);
  - `check_fukaya_probe.py` (6 checks) and `fukaya_scan.py`: [proposition 10.9](#), [observation 10.10](#).
  - `check_tower_horns.py` (9 checks): [theorem 11.2](#), [corollary 11.3](#), and [propositions 11.4](#) and [11.5](#).

## A Corrections to the first draft

The items below record the remaining corrections to the first draft.

1. *Candidate disc potential*.  $W_\beta(f) = \sum_g \int_{\text{Cpl}(f,g)} e^{-\beta c(\alpha)} d\nu$  sums over a continuum of 1-cells  $g$  and is not defined as written. Its Laplace limit  $\min_{g,\alpha} c(\alpha)$  is always 0, attained at  $g = f$ ,  $\alpha = \Delta_f$ , so it carries no information. The toric disc potential of  $X_{f,g}$  is the correct object ([section 7](#)).
2. *Symplectic structure*. A symplectic form is antisymmetric and the Fisher metric is symmetric, so the restriction of the one cannot be the other. The correct statement is [proposition 6.5](#): the Fisher metric is twice the canonical Kähler metric on the real section of  $X_{f,g}$ .

Table 1: Withdrawn identifications, the reason, and what is claimed instead.

| Withdrawn                                           | Why                                                                                                                                                                                                                                                | Claimed instead                                                                                                                                                                          |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disc potential $W$ as a central charge              | A central charge is an additive map $K_0 \rightarrow \mathbb{C}$ ; $W$ is a function on a torus                                                                                                                                                    | The central charge $Z_\omega$ of the product-coupling class ( <a href="#">proposition 8.1</a> ). $W$ enters only through the oscillatory integrals $\int e^{-W/z}$ of (Q3)               |
| Committed states as a heart                         | A heart is an abelian category; committed states are vertices of a polytope                                                                                                                                                                        | The tilted heart of $\text{Coh } X_{f,g}$ ( <a href="#">proposition 8.1</a> ); committed states are torus-fixed points                                                                   |
| Obstruction tower as a Harder–Narasimhan filtration | An HN filtration is a chain of subobjects of one object, with decreasing phases; the tower assigns numbers to a claim                                                                                                                              | A convex obstruction theory with certificates; its monotonicity [2, Thm. 10.5] and <a href="#">proposition 11.5</a> are not a filtration                                                 |
| Hallucination as instability                        | It depended on the three identifications above                                                                                                                                                                                                     | Hallucination as obstructed lifting [2, Def. 10.17]; unfillable horns are instances ( <a href="#">theorem 11.2</a> )                                                                     |
| $F_2 = \text{Hess } W$ ; Picard–Lefschetz formula   | $F_2$ is bilinear on composable couplings with values in circulations; $\text{Hess } W$ is a quadratic form at a critical point of a function on a torus. The cited check compared 0.1152 with 2, and no vanishing cycle was linked to composition | $F_2 = \varepsilon b_2$ , a Boolean cumulant ( <a href="#">theorem 5.2</a> ). $\text{Hess } W$ matches the Fisher metric only at the monotone point ( <a href="#">proposition 10.9</a> ) |
| The $m_3$ conjecture                                | The whiskering defect is the second component of a functor, not a ternary composition (item 5 below)                                                                                                                                               | Whiskering is a two-component $A_\infty$ -functor ( <a href="#">theorem 5.2</a> ); higher components are <a href="#">question 9.4(i)</a>                                                 |
| 22 chambers in the $2 \times 3$ family              | A coordinate-order bug in the vertex code                                                                                                                                                                                                          | 18 chambers in 5 orbits [4, Thm. 3.1]                                                                                                                                                    |
| Cycle counts of $K_{n,n}$                           | The formula undercounted by a factor $k$ ; facets are cells, not cycles                                                                                                                                                                            | $N_{2k} = \binom{n}{k}^2 k!(k-1)!/2$ ; $K_{4,4}$ has 204 cycles [4, App. A]                                                                                                              |
| Bridgeland’s construction for all surfaces          | [7] treats K3 surfaces only                                                                                                                                                                                                                        | Arcara–Bertram [3] for any smooth projective surface and ample class                                                                                                                     |

3. *Critical points.* “Critical points of  $\pi: E \rightarrow \text{Kl}(\mathbf{D})$ ” is undefined, since  $\pi$  is a functor, not a smooth map. The replacement is the special fibre over the product coupling in the monotone cases, and the walls of [definition 6.3](#) as the locus where the family degenerates.
4. *Monodromy and Dehn twists.* The family  $(f, g) \mapsto X_{f,g}$  is constant on chambers and changes by toric birational modifications across walls ([proposition 6.6](#)); no vanishing cycle or Dehn twist has been identified. [Conjecture 9.2](#) restates the question.
5. *The  $m_3$  conjecture.* Gluing is strictly associative ([theorem 2.3\(i\)](#)), so each hom-category, regarded as an  $A_\infty$ -category, has  $m_k = 0$  for  $k \geq 3$ . The whiskering defect compares  $u \cdot (\beta \cdot \alpha)$  with  $(u \cdot \beta) \cdot (u \cdot \alpha)$ . Its inputs are two 2-cells and a 1-cell, and it measures the failure of whiskering to be a functor. In  $A_\infty$  language that is the second component  $\mathcal{F}_2$  of an  $A_\infty$ -functor, not a ternary composition (now [theorem 5.2](#)).
6. *Homotopy category.* The graded shadow has hom-sets  $[d(f, g), \infty)$  under addition. These are not

the degree-zero cohomology of cochain complexes, so the shadow is not a homotopy category of an  $A_\infty$ -category without further structure.

7. *Fisher versus Fisher–Rao*. These are the same metric ([proposition 6.1](#)).
8. *References*. The companion tower is now version 9; its posted earlier version has the title “... and Weak Distributive Laws”, a claim withdrawn in the revisions.

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